

CHAPTER 10

Wave Optics

Intensities of Maxima and Minima

1. The interference pattern is obtained with two coherent light sources of intensity ratio n . In the interference pattern, the ratio $\frac{I_{\max} - I_{\min}}{I_{\max} + I_{\min}}$ will be: (2016 - II)

a. $\frac{\sqrt{n}}{(n+1)^2}$ b. $\frac{2\sqrt{n}}{(n+1)^2}$ c. $\frac{\sqrt{n}}{n+1}$ d. $\frac{2\sqrt{n}}{n+1}$

Interference and Young's Double Slit Experiment

2. Two slits in Young's double slit experiment are 1.5 mm apart and the screen is placed at a distance of 1 m from the slits. If the wavelength of light used is 600×10^{-9} m then the fringe separation is (2024 Re)

a. 4×10^{-5} m b. 9×10^{-8} m
c. 4×10^{-7} m d. 4×10^{-4} m

3. Interference pattern can be observed due to superposition of the following waves:

A. $y = a \sin \omega t$ B. $y = a \sin 2\omega t$
C. $y = a \sin(\omega t - \phi)$ D. $y = a \sin 3\omega t$

Choose the correct answer from the options given below. (2024 Re)

a. B and C b. B and D
c. A and C d. A and B

4. If the monochromatic source in Young's double slit experiment is replaced by white light, then (2024)

- a. there will be a central bright white fringe surrounded by a few coloured fringes.
b. all bright fringes will be of equal width.
c. interference pattern will disappear.
d. there will be a central dark fringe surrounded by a few coloured fringes.

5. For Young's double slit experiment, two statements are given below: (2023)

Statement I: If screen is moved away from the plane of slits, angular separation of the fringes remains constant.

Statement II: If the monochromatic source is replaced by another monochromatic source of larger wavelength, the angular separation of fringes decreases.

In the light of the above statements, choose the correct answer from the options given below:

- a. Statement I is false but Statement II is true
b. Both Statement I and Statement II are true
c. Both Statement I and Statement II are true
d. Statement I is true but Statement II are false

6. If the screen is moved away from the plane of the slits in a Young's double slit experiment, then the: (2022 Re)

- a. linear separation of the fringes decreases
b. angular separation of the fringes increases
c. angular separation of the fringes decreases
d. linear separation of the fringes increases

7. In a Young's double slit experiment, a student observes 8 fringes in a certain segment of screen when a monochromatic light of 600 nm wavelength is used. If the wavelength of light is changed to 400 nm, then the number of fringes he would observe in the same region of the screen is (2022)

a. 12 b. 6 c. 8 d. 9

8. In Young's double slit experiment, if the separation between coherent sources is halved and the distance of the screen from the coherent sources is doubled, then the fringe width becomes: (2020)

a. Half b. Four times c. One-fourth d. Double

9. Two coherent sources of light interfere and produce fringe pattern on a screen. For central maximum, the phase difference between the two waves will be. (2020-Covid)

a. π b. $3\pi/2$ c. $\pi/2$ d. Zero

10. In a double slit experiment, when light of wavelength 400 nm was used, the angular width of the first minima formed on a screen placed 1 m away, was found to be 0.2° . What will be the angular width of the first minima, if the entire experimental apparatus is immersed in water? ($\mu_{\text{water}} = 4/3$) (2019)

a. 0.266° b. 0.15° c. 0.05° d. 0.1°

11. In Young's double slit experiment the separation d between the slits is 2 mm, the wavelength λ of the light used is 5896 Å and distance D between the screen and slits is 100 cm. It is found that the angular width of the fringes is 0.20° . To increase the fringe angular width to 0.21° (with same λ and D) the separation between the slits needs to be changed to (2018)

a. 2.1 mm b. 1.9 mm c. 1.8 mm d. 1.7 mm

12. Young's double slit experiment is first performed in air and then in a medium other than air. It is found that 8th bright fringe in the medium lies where 5th dark fringe lies in air. The refractive index of the medium is nearly: (2017-Delhi)

a. 1.59 b. 1.69 c. 1.78 d. 1.25

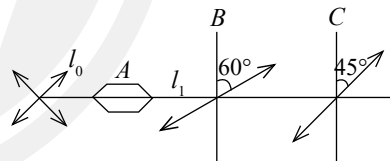
13. The intensity at the maximum in a Young's double slit experiment is I_0 . Distance between two slits is $d = 5\lambda$, where λ is the wavelength of light used in the experiment. What will be the intensity in front of one of the slits on the screen placed at a distance $D = 10d$? (2016 - I)
 a. I_0 b. $\frac{I_0}{4}$ c. $\frac{3}{4}I_0$ d. $\frac{I_0}{2}$
14. Two slits in Young's experiment have widths in the ratio 1 : 25. The ratio of intensity at the maxima and minima in the interference pattern, $\frac{I_{\max}}{I_{\min}}$ is: (2015 Re)
 a. 4/9 b. 9/4 c. 12/149 d. 49/121
15. In the Young's double-slit experiment, the intensity of light at a point on the screen where the path difference is λ is K , (λ being the wavelength of light used). The intensity at a point where the path difference is $\lambda/4$, will be: (2014)
 a. K b. $4K$ c. $K/2$ d. Zero
16. In Young's double slit experiment, the slits are 2 mm apart and are illuminated by photons of two wavelengths $\lambda_1 = 12000 \text{ \AA}$ and $\lambda_2 = 10000 \text{ \AA}$. At what minimum distance from the common central bright fringe on the screen 2 m from the slit will a bright fringe from one interference pattern coincide with a bright fringe from the other? (2013)
 a. 4 mm b. 3 mm c. 8 mm d. 6 mm

Diffraction of Light From a Narrow Slit

17. In a diffraction pattern due to a single slit of width a , the first minima is observed at an angle 30° when light of wavelength $5000 \text{ \AA}$ is incident on the slit. The first secondary maximum is observed at an angle of: (2016 - I)
 a. $\sin^{-1}\left(\frac{1}{4}\right)$ b. $\sin^{-1}\left(\frac{2}{3}\right)$
 c. $\sin^{-1}\left(\frac{1}{2}\right)$ d. $\sin^{-1}\left(\frac{3}{4}\right)$
18. A linear aperture whose width is 0.02 cm is placed immediately in front of a lens of focal length 60 cm. The aperture is illuminated normally by a parallel beam of wavelength $5 \times 10^{-5} \text{ cm}$. The distance of the first dark band of the diffraction pattern from the center of the screen is: (2016 - II)
 a. 0.20 cm b. 0.15 cm c. 0.10 cm d. 0.25 cm
19. For a parallel beam of monochromatic light of wavelength ' λ ', diffraction is produced by a single slit whose width ' a ' is of the order of the wavelength of the light. If ' D ' is the distance of the screen from the slit, the width of the central maxima will be: (2015)
 a. $\frac{D\lambda}{a}$ b. Da/λ c. $12D/a$ d. $\frac{2D\lambda}{a}$
20. In a double slit experiment, the two slits are 1 mm apart and the screen is placed 1 m away. A monochromatic light of wavelength 500 nm is used. What will be the width of each slit for obtaining ten maxima of double slit within the central maxima of single slit pattern? (2015)
 a. 0.1 mm b. 0.5 mm c. 0.02 mm d. 0.2 mm
21. At the first minimum adjacent to the central maximum of a single-slit diffraction pattern the phase difference between the Huygens's wavelength from the edge of the slit and the wavelet from the mid point of the slit is: (2015 Re)
 a. $\frac{\pi}{8}$ b. $\frac{\pi}{4}$ c. $\frac{\pi}{2}$ d. π
22. A beam of light of $\lambda = 600 \text{ nm}$ from a distant source falls on a single slit 1 mm wide and the resulting diffraction pattern is observed on a screen 2 m away. The distance between first dark fringes on either side of the central bright fringe is: (2014)
 a. 1.2 cm b. 1.2 mm c. 2.4 cm d. 2.4 mm
23. A parallel beam of fast moving electrons is incident normally on a narrow slit. A fluorescent screen is placed at a large distance from the slit. If the speed of the electrons is increased, which of the following statements is correct? (2013)
 a. The angular width of the central maximum will be unaffected
 b. Diffraction pattern is not observed on the screen in the case of electrons
 c. The angular width of the central maximum of the diffraction pattern will increase
 d. The angular width of the central maximum will decrease

Polarization Phenomenon, Law of Malus and Brewster's Law

24. A beam of unpolarized light of intensity I_0 is passed through a polaroid A , then through another polaroid B , oriented at 60° and finally through another polaroid C , oriented at 45° relative to B as shown. The intensity of emergent light is: (2024 Re)



- a. $\frac{I_0}{16}$ b. $\frac{I_0}{4}$ c. $\frac{I_0}{2}$ d. $\frac{I_0}{32}$
25. An unpolarised light beam strikes a glass surface at Brewster's angle. Then (2024)
 a. both the reflected and refracted light will be completely polarised.
 b. the reflected light will be completely polarised but the refracted light will be partially polarised.
 c. the reflected light will be partially polarised.
 d. the refracted light will be completely polarised.
26. After passing through a polariser a linearly polarised light of intensity I is incident on an analyser making an angle of 30° with that of the polariser. The intensity of light emitted from the analyser will be: (2022 Re)
 a. $\frac{2I}{3}$ b. $\frac{I}{2}$ c. $\frac{I}{3}$ d. $\frac{3I}{4}$
27. The Brewsters angle i_b for an interface should be : (2020)
 a. $30^\circ < i_b < 45^\circ$ b. $45^\circ < i_b < 90^\circ$
 c. $i_b = 90^\circ$ d. $0^\circ < i_b < 30^\circ$

28. Unpolarised light is incident from air on a plane surface of a material of refractive index ' μ '. At a particular angle of incidence ' i ', it is found that the reflected and refracted rays are perpendicular to each other. Which of the following options is **correct** for this situation? (2018)

a. $i = \sin^{-1}\left(\frac{1}{\mu}\right)$

- b. Reflected light is polarised with its electric vector perpendicular to the plane of incidence

- c. Reflected light is polarised with its electric vector parallel to the plane incidence

d. $i = \tan^{-1}\left(\frac{1}{\mu}\right)$

29. Two Polaroids P_1 and P_2 are placed with their axis perpendicular to each other. Unpolarised light I_0 is incident on P_1 . A third polaroid P_3 is kept in between P_1 and P_2 such that its axis makes an angle 45° with that of P_1 . The intensity of transmitted light through P_2 is: (2017-Delhi)

a. $\frac{I_0}{4}$

b. $\frac{I_0}{8}$

c. $\frac{I_0}{16}$

d. $\frac{I_0}{2}$

Answer Key

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|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| 1. (d) | 2. (d) | 3. (c) | 4. (a) | 5. (d) | 6. (d) | 7. (a) | 8. (b) | 9. (d) | 10. (b) |
| 11. (b) | 12. (c) | 13. (d) | 14. (b) | 15. (c) | 16. (d) | 17. (d) | 18. (b) | 19. (d) | 20. (d) |
| 21. (d) | 22. (d) | 23. (d) | 24. (a) | 25. (b) | 26. (d) | 27. (b) | 28. (b) | 29. (b) | |